

Food Waste Prediction Using Machine Learning Models

Vanshak Bhardwaj

Department of Computer Science
and Engineering

Chitkara University Institute of
Engineering and Technology,

Chitkara University Rajpura,

Punjab, India

vanshak942.be22@chitkara.edu.in

Gurpreet Singh

Department of Computer Science
and Engineering

Chitkara University Institute of
Engineering and Technology,

Chitkara University Rajpura,

Punjab, India

gurpreet.1309@chitkara.edu.in

Abstract- In fact, one out of three food items manufactured throughout the world remains unsold, totalling well over a billion tons per annum. The most part of it goes missing in private homes, which generate almost two-thirds of the food waste after consumption. According to estimates, in India alone yearly household waste amounts to 78 million tons, which corresponds to about fifty kilograms per individual annually. Although plenty of food gets spoiled, many inhabitants face inadequate access to the necessary nutrition, not to mention financial implications. A computational learning algorithm identifies the pattern of food wastage by examining different metrics such as ratio of utilization to purchases, frequency of spoilage, factors involved in the process, and so forth: season, occasion, demographics, and regularities. Predictive analytics helps minimize losses by fine-tuning the inventory orderings, making dynamic changes in menus, and so on. Instead of reacting to the situation, the organization manages to foresee the necessity and apply a more intelligent forecast into their routine. There exist several methods; for example, a certain approach is based on previous statistics and predicts demands of the future, whereas another method identifies patterns of disposal in relation to the waste generation. Some approaches use live data from smart sensors installed in the kitchen and pantry. As for algorithms, Random Forest and LSTM have proven themselves to be the best performers and yield precision over 90% in waste reduction from 14% to 52%. Still, the problem includes dealing with imprecise data, transferring a model to another setting, and generalizing from experiments to the practice. Further research will need to concentrate on local context and specificity of an area..

Keywords: Food waste prediction, machine learning, demand forecasting, Random Forest, LSTM, sustainability

I. INTRODUCTION

From farms to forks, uneaten food piles up across every stage. When crops rot before harvest or spoil during transport, that is loss piling up behind the scenes. Shops toss imperfect produce; restaurants serve oversized portions; homes throw away leftovers - these habits fuel household-level waste. Rotting scraps buried underground pump gases into the air, helping warm the planet. That stashed trash accounts for a chunk of human-driven emissions, roughly between eight and ten out of every hundred tons. All that decay drains precious resources too - gallons soaked into vanished meals, fields left idle, power spent for nothing. A UN target pushes for smaller footprints: slash

store and home waste by half within this decade. Trim supply route losses just as hard. Aim set. Deadline ticking.

Old ways of handling trash usually guess amounts or use rough estimates, leading to too much waste or poor preparation. Because it spots hidden trends in messy real-world information, machine learning does better at spotting complex patterns over time. Instead of just adding numbers, systems now predict food needs in cafeterias, track how families toss leftovers, even catch rot using camera images or device readings. When waste runs high - say, homes or eateries across India, where nearly half of fresh produce spoils before eating - smarter tools come into play. Think: bins that know when they'll fill up, ordering plans adjusted daily based on predicted meals, software helping plan dinners with less throwaway. The work gathered here doesn't repeat old studies; rather, it lines up methods side by side, weighs their strengths, outlines what actually matters once you start putting models to use.

II Datasets and Features

Good machine learning needs clean information. Typical places people get it are these:

Open data sources include Kaggle collections such as the "Messy Food Waste Prediction Dataset," useful when estimating kilograms of waste post-cleaning. National figures appear in reports by FAO, UNEP, and JRC - these span trash from homes, stores, and eateries. Information drawn from past sales records helps too. Weather feeds add context over time. So do public schedules tied to holidays or gatherings. Population details shift the picture further.

Starting off, weight measurements come from load cells inside containers. Moving on, bin fullness gets tracked using real-time **sensor** feedback. Temperature readings tag along as part of the monitoring mix. Images tagged to items help sort them through visual recognition. Each piece feeds into the system without needing extra steps. Patterns in what people buy show up through surveys. Behaviour clues come from how folks act over time. Attitudes mix with income levels to shape choices. Past buys hint at future picks when viewed closely.

Such characteristics commonly include:

The blending of weekdays with weekends into one stream. Seasonal transition - one season gradually turning into another. Holidays appearing unexpectedly. The time of day changing during the blink of an eye. Perception changes depending on the time of day.

From the old days, sales performance guided decisions for now. They attended events when specific meals were served. Promotions brought additional attendees at last year's event. Things that proved effective before became a part of the current menu offerings.

- Environmental: Weather conditions, storage settings.
- Contextual: Venue type, customers' demographics, economic factors.
- Operational: Current inventory, serving portions, batch sizes.

Unusual gaps in data may appear from time to time. Uncommon events of the waste sort affect the class balance more than they should. Unnecessary noise pops up when datasets turn into something unclear unexpectedly. Privacy concerns arise due to personal information about homeowners included. The cleansing step usually takes place first. Normalization balances scales to make the next steps more accurate. Handcrafted features like historical observations assist machine learning algorithms a lot. Additional instances occur using augmentation techniques afterward.

Three. Ways computers learn from data using special rules and systems

3.1 Regression and Forecasting Models

Amount of waste? Hardly relevant. Levels of demand will be evident through consumption patterns. Surpluses will be revealed with the help of data dynamics - observe growth patterns. Forecasting will work where history tends to repeat itself. Data analysis shows how much may happen if done properly. The starting point is linear regression models, which do reasonably well in cases where trends are easy to track. However, fails when complex curves occur in the middle of processes. Traditional models won't help at all in instances of unpredictable relationships. Attempts to tweak them have been made but didn't deliver much improvement either. Tree ensembles, in most cases, provide better accuracy as they deal well with various data types. Contrary to traditional models that rely on single models, ensemble solutions involve multiple decision trees. Among all methods for ranking input features, Random Forest stands out. Furthermore, algorithms such as XG Boost successfully withstand data noise issues without sacrificing performance. When it comes to logistics operations and scheduling, for example in a restaurant kitchen, these models prove to be very stable and consistent despite noisy inputs. Performance metrics do not vary too drastically.

One approach to tracking temporal dependencies is an ARIMA model - a classic statistical method. However, there are other options that use deep neural networks, such as LSTM or GRU. These can be used to identify long-term dependencies as they can track trends in trash generation through periods. Although it is another approach to the task, both seek to identify trends in data. There isn't a universal model for any cycle, but some adjust to unpredictable peaks quite well. The important issue is the ability to account for delay between events.

3.2 Classification Models

- For waste level classification (high waste / low waste) and waste type classification:

- Decision Trees, SVM, KNN, MLP.
- Ensemble methods also work quite effectively here.

3.3 Deep Learning Meets Hybrid Methods

- Convolutional neural networks for detecting spoiled food in photos.
- IoT-Machine learning hybrid systems for prediction purposes.

Inventory management strategy may employ reinforcement learning to adjust order size based on previous shortages or surplus of product. This strategy will learn to find balance in time by observing consequences. Thus, there will be no pre-calculated fixed rules but rather a reaction to changes in external circumstances such as a peak in demand. After several weeks of training, patterns emerge minimizing waste while meeting customer requirements.

IV. LITERATURE REVIEW

Lately, studies into machine learning for tackling food waste have picked up speed across various parts of the supply chain. At restaurants and shops, guessing what customers will buy helps stop too much food being made. Looking at recent work, Rodrigues and team in 2024 tested several ways to predict meals needed each day, noticing that Random Forest along with LSTM systems outperformed older techniques, cutting possible waste between 14% and 52% by aligning cooking more closely with actual eating patterns. Because they resist noise well while spotting complex links among variables - such as how many guests show up, menu choices, weekday rhythms, public holidays, or store setting - methods like Random Forest tend to perform reliably when blending these inputs together. Starting with smart trash containers, one project sends sensor readings straight to online systems. Instead of simple connections, these devices rely on weight detectors or space monitoring tools that update live. Once uploaded, artificial intelligence sorts the material type or guesses when bins will overflow - results landing near 93% correct. When factories plan output, machines step in to handle guesswork around supply shifts or customer needs. Those setups hit about 92% precision forecasting garbage volume. At home, forecasts lean on questionnaires or habits people show over time. In a recent case, researchers used decision paths and brain-inspired algorithms after grouping shoppers by pattern, pulling insights from polls tied to farm-to-table cycles. Elsewhere, another team reached nearly perfect scores - climbing to 99% in certain tests - by training methods such as random picks, support vectors, and boosted groups on answers collected earlier, which then shaped advice for lawmakers.

Computers that see - like those using CNNs - spot spoilage by studying pictures. Temperature, moisture, or acidity readings also help detect decay early on. These signals often pair with tools like XG Boost when tracking gases from rotting matter. Stores and eateries already apply such tech to cut down waste. Results show drops between 14.8% and 30%, thanks to smarter guesses about stock needs. Yet in places like India, habits around

food differ widely across regions. Seasons shift what grows where, making one-size-fits-all models less effective. City versus village access adds more complexity too. Public data there? Still thin when stacked against worldwide collections.

Still, the bulk of effort targets big industries, yet little reaches smaller, hands-on setups. This project takes those concepts, shaping them into something clear and down-to-earth.

V. Model Evaluation

Errors often show up as MAE or RMSE when predicting numbers, while MAPE gives a percent view of mistakes. For sorting items into groups, Accuracy tells how many are right overall, Precision checks false alarms, Recall measures misses, and F1 balances both. Real-world impact might track less waste by weight or fewer carbon emissions saved over time. Splitting time-based data carefully stops leaks during testing. Tuning settings happens through grid checks or smart guessing methods. Tools such as SHAP reveal how forecasts form, which builds confidence when sharing results in meal-related operations.

VI. Experimental Framework (Conceptual Implementation)

A typical pipeline:

1. Data Collection: Aggregate historical waste/sales data, enrich with external features.
2. Outliers get adjusted first. Missing numbers filled next, using reasonable estimates. Categories turned into numbers after that. Feature sizes matched last, through scaling methods.
3. Start by picking features through correlation checks or stepwise removal methods. Moving on, divide data carefully - keep time sequence intact - into three parts like seventy, fifteen, then another fifteen.
4. Try different learners such as forest-style trees, boosted gradients, or memory-based networks. Each choice shapes how the system learns patterns over steps.
5. Hold-back data checks how well it works. Instead of just adding up past numbers, see if predictions beat a flat guess every day. Results show whether smarter methods pull ahead or fall short.
6. Live updates show right inside tools people already use, like warnings saying cut next day's supply amount by one fifth. Most times, decision trees handle number grids well enough without slowing things down. When patterns come in order, like daily trash amounts piling up, memory-based models adapt better over stretches of time.

VII. Challenges and Limitations

For example, collecting precise records about the volume of household waste is almost impossible, even when talking about some areas of India. Patterns from a specific setting may be irrelevant in other locations. Eating behaviour may significantly vary from household settings to a restaurant's kitchen. Data from restaurant services is irrelevant for understanding consumer behaviour in the home. Weather conditions affect food consumption and preparation. The algorithms fail to account for cultural differences, despite them impacting consumption habits greatly.

The deep-learning models are more accurate but leave us unaware of what is happening inside. Despite the ability of deep-learning solutions to make more accurate predictions, the models remain opaque and do not allow understanding the way they make particular predictions.

Technology that is already ancient and is used in small restaurants or rural areas may hinder the implementation of the solutions. Moreover, the deployment of modern Internet-powered sensors would be expensive and unaffordable. Old technology does not interact with new technology well. Installation and integration of new equipment require money. The problem of cash flow is an issue as costs grow quickly, prompting some business owners to avoid upgrading their equipment. They continue using their familiar old technology. Sometimes, it could be argued that prediction per se does not matter as much. It is crucial to understand the root cause of the behaviour and fix it. Consumption patterns influence the decision-making process. Advertising affects choices. Unless the driving forces of the decision-making process are addressed, the change will be hardly significant.

Sometimes, festivals occur and natural disasters strike, affecting the current state of things. Machine learning approaches must be able to learn while operating during the period of fluctuations. It is important to take unpredictable events into account to ensure flexibility of the algorithms.

Sometimes, the reliance on previous trends leads to failure as there may be other factors that affect the process significantly, such as governmental action and changes in public opinion. Over-reliance on the past trends may prevent the identification of important events that change things dramatically. Changes in consumer behaviour do not happen instantly because of governmental regulations or increased awareness.

VIII. Implications for Sustainability and Policy

Because predictions improve, efforts toward SDG 12.3 get stronger through focused actions like changing prices based on demand, linking surplus food to those who need it - such as scaling up programs similar to India's "Save Food Share Food" - streamlining how goods move from farm to table, while also shaping public awareness using insights into human behaviour. In Indian communities, pairing machine learning tools with current farming support systems or urban development plans may sharply cut down loss after harvest and less waste at home, opening room to feed more people reliably. Less rotting food means fewer methane fumes rise into air, helping nature stay balanced along with saving water and energy used in production. Money flows better when spoiled items drop off ledgers - billions might be preserved - and fresh ways of earning appear by turning scraps into useful things such as compost or gas for fuel.

IX. Future Directions

Tabled data accompanies the images that accompany them. The menu descriptions accompany the pictures of the dishes. Data acquired through sensors also follows. There is an integration of textual elements with visual ones. It involves the pairing of pictures with tabulated data. Descriptions are included with sensor input. The various types of data are overlapping - tables with camera shots with written texts.

Data-free training of models - it keeps data within its respective kitchen/home. Information does not move around while one site can learn from another through quiet paths.

- Explainable AI (XAI): Increase the trustworthiness and practical nature of AI.

The date of events that guide systems' responses. Monsoons affect sensor activity. Local characteristics weigh heavier than advanced technology. Adaptation of simple technology based on regional cycles. Natural climate conditions affecting day-to-day activities. Awareness informed by the practices of the community. Sensory activity adapted to real-time cycles of natural processes. Design customized according to regional needs.

- End-to-End Optimization: Combine predictive and prescriptive analytics (e.g., applying reinforcement learning to inventory ordering policies). Through monitoring trends over months/years, it becomes evident what wastes need reducing and where. In highly-populated countries like India, this could lead to faster change simply due to more people being involved. Collaboration between coders analysing numerical data and suppliers of grocery stores, as well as government and decision scientists, tends to be more successful. Success often results from the collaboration of mismatched groups.

X. CONCLUSION

In the world of logistics for supplies, machine learning comes in handy in making accurate forecasts of future food demands before they even occur. Predictions using decision tree algorithms or neural networks are capable of reducing overproduction and food wastage significantly – up to ten percent when used effectively. In all cases, the success of any such endeavour will depend on the accuracy and relevance of the data provided. Machine learning cannot work magic unless the collected data is properly connected to the actual process. It is

important to point out that in the open market, machine learning helps reduce food waste, thus improving systems efficiency while saving the environment. The modern rate of data generation provides more opportunities to create highly sophisticated algorithms for forecasting food transportation. This involves the synthesis of knowledge from diverse sources. Consequently, scientists have created a forecast model that changes the course of food transportation to some degree. As the change occurred, India, with its vast food waste problem, became an immediate target.

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